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Blockchain in Agro-Industrial Complex

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Abstract. The article reviews the available projects and programs using digital technologies and the results of their adaptation in agriculture and specifically in the agro-industrial complex. Tokenization makes any production sector attractive and transparent for investment. The blockchain enables an investor to understand and estimate where the money is invested and what the result will be, how large the final profit will be and whether there will be an additional investment income from the agribusiness. Most of the initial efforts in blockchain in agribusiness are trackable achievements. The authors point out that it is possible to create an efficient supply chain that provides full product serviceability and improves the transactions efficiency by using modern digital technologies. It is about improving margins by reducing bureaucracy and supporting continuity of business problem-solving with blockchain. In doing so, blockchain provides all stakeholders with a common source of reliable information for the agro-industrial and logistics chain.

1. Introduction

It is the current reality that the use of digital financial assets as a means of payment and the application of blockchain technology is becoming an absolutely effective means of saving money and reducing the risk of fraud, including in the agricultural sector. As we know, blockchain is a chain of blocks, but it is not just a chain. It has a strict sequence defined by complex cryptographic functions [1]. Blocks are data about transactions, deals and contracts within the system, represented in cryptographic form. All blocks are arranged in a logical and continuous chain, i.e. they are sequentially connected to each other. To add (record) a new block, one has to check the sequence of previous blocks [2].

Modern computers are extremely powerful and can calculate thousands of hashes per second. Therefore, there is a high probability of forging a block and making changes to all subsequent blocks to make the blockchain valid and working again. To prevent such cases, the Proof of Work algorithm is



built into the blockchain. Its essence is to find a hash of a function whose result starts with a certain number of zeros. This mechanism slows down the creation of new blocks and protects against DDoS attacks [3, 4]. In the case of bitcoin, it takes about 10 minutes to calculate a new block and add it to the overall block sequence of the network. Proof of Work greatly complicates the possibility of block counterfeiting, because once a block is counterfeited, all subsequent blocks must also be recalculated using Proof of Work. The joint use of this mechanism with block hashes is the basis of blockchain security in absolutely all spheres of human activity [5].

Recording and registering transactions in agribusiness play a key role in ensuring legal relations in agriculture. However, existing recording and registration systems often suffer from problems such as inefficient data recording, high costs, and the possibility of data tampering [6]. In recent years, blockchain technology has been widely recognized for its ability to provide reliable, trustworthy, transparent and fixed, non-modifiable data records. Therefore, it is relevant to investigate the possibility of applying blockchain technology to record and register all kinds of activities in related industries [7].

Blockchain technology is based on a number of key principles that ensure its reliability and transparency. One of the key principles is decentralization, i.e., no central control or data storage. Blockchain consists of a network of nodes that interact with each other and maintain a common state of the system [8]. The second important principle is immutability. Each transaction in the blockchain is written as a block, which is then linked to the previous block using a hash function. This creates a chain of blocks where each block contains information about the previous block, making it very difficult and costly for the customer to change past records [9]. Another principle is transparency. All transactions on the blockchain are public and available for viewing by all participants in the network. This promotes trust and openness in the system, as anyone can check the transaction history.

Financial organizations, universities and research centers are increasing the number of blockchain patents, with China and the United States leading the way. According to the results of the QIWI Blockchain Technologies survey, 2020 was the year of experiments with the use of blockchain technologies in the largest electric companies and associations. For banks, it is also the time of decisive actions: transition to new technologies, primarily digital; consolidation through mergers; cost reduction, including by outsourcing some secondary tasks. Since 2020, many countries have gone far ahead in their development of digital technologies and have successfully used blockchain in various spheres of production.

2. Materials and Methods

The term “blockchain” means a state of the economy and agro-industrial sector of the country, which, while preserving and improving the environment, regardless of external and internal conditions, allows the country’s population to consistently receive ecologically safe and healthy food products at affordable prices, in amounts not lower than scientifically substantiated norms.

The theoretical research was based on academic works, statistical studies on the food security digitalization, and analytical materials on the production sector development. The methodological basis of the study was a combination of general scientific methods such as abstraction, analysis and synthesis, induction and deduction and others.

3. Results and Discussion

According to the search queries “digital economy”, “cryptocurrency”, “blockchain”, “food security”, “analysis and assessment of risks and threats to financial and economic security”, etc., it is possible to select a number of recent publications by such authors as A. V. Babkin, M. L. Vartanova, E. V. Drobot, E. A. Zotova, M. V. Mazurovskiy, T. V. Kuladzhi, L. V. Kozlova, I. G. Ushachev and others [10, 11, 12]. Among the studies in this area listed and studied by us, special attention should be paid to the works by M. L. Vartanova [13], which identify the problems of food security and ways to solve them, A. V. Babkin [10, 11], who analyzed the development and regulation of cryptocurrencies, taking into account foreign and Russian experience, and scientific works by I. G. Ushachev [12], devoted to the challenges and prospects for the development of Russia’s agricultural sector under modern conditions.

This year Russia will undergo a number of significant revolutionary changes in the agro-industrial complex. In particular, it was decided to create a unified digital platform and put it into operation. The purpose of the platform is to collect and store information on the entire agro-industrial and fishery complex of Russia. The platform is to work solely on Russian software, which is still being developed. Such platforms are costly in production and maintenance; they are complex technological processes requiring efforts of many professional communities. This confirms the constant desire of the state to increase the effective use of available resources and digital technologies in various sectors of economic activity.

To ensure data security and privacy when using blockchain technology, one may emphasize three main points:

1) Cryptographic data protection mechanisms play an important role in securing data on the blockchain. They include hashing, digital signatures and encryption. Hashing is used to create a unique identifier for each block to detect any changes in the data. Digital signatures provide authentication and data integrity, while encryption ensures confidentiality of information transmission. The combination of these cryptographic mechanisms protects data from unauthorized access and tampering [14];

2) Access and privacy management in blockchain is an important aspect when using blockchain in accounting systems. Blockchain can be configured for various access levels, with certain data to be accessed by certain network participants only. This helps to ensure the confidentiality of personal information or trade secrets that must not be available to all participants [15];

3) Reducing the risk of fraud and falsification of documents is one of the significant advantages of blockchain in agro-industrial systems. Blockchain ensures that past records cannot be altered without consent of the majority of network participants. This allows achieving a high degree of data reliability and integrity, and prevents possible manipulation or forgery of documents [16].

A comprehensive blockchain system for agro-industrial complex should be launched and implemented in 2025. There is not much time left, but a number of issues need to be addressed now. In particular, it is necessary to create an effective blockchain in the agrarians' operating system, which will ensure food security of the Russian Federation for many years [17, 18].

It is planned to launch the digitalization of agro-industrial and fishery services in the Russian Federation in several stages, as this will increase its efficiency. It seems realistic to introduce digital format in 50% of all processes in the agro-industrial sector in 2026, and further the pace will only increase. It is planned to digitize the entire agricultural system by 2028.

The goal is to introduce artificial intelligence and digital technologies not only at a rapid pace, but also with the highest possible quality into all spheres of human life, including production activities [19, 20]. These resources will reduce the time of services rendering and improve their quality. The important directions are: increasing yields, optimizing costs, finding new channels for marketing products and, among others, new supply channels. All benefits for agriculture should yield a concrete result. As for the regulatory system within the current legislation, the draft law on creating an information system of digital services in the agro-industrial complex was submitted to the State Duma of the Russian Federation by the government back in December 2021. In May 2022, the law was adopted, and in July of the same year the procedure for the operation of the new information system of digital services in the agro-industrial complex was determined. Assumingly, the system will allow interested entities to apply for state aid electronically, and help representatives of state authorities to assess the effectiveness of state regulation using the established system of data collection and reporting.

4. Conclusion

The following benefits of blockchain technology in the agro-industrial complex were highlighted and a list of blockchain benefits was compiled. First, blockchain ensures the reliability and integrity of data. Changing records in the blockchain requires the consent of the majority of network participants, making data manipulation virtually impossible. Second, blockchain improves the efficiency of record-keeping and registration processes. Due to automation and electronic transfer of information, the time to perform procedures is reduced and the bureaucratic load on the public authorities is lifted [21]. This is especially

important in the agricultural sector, where fast and accurate transactions can be critical. In addition, blockchain promotes transparency and improved trust in the contract accounting system within the agro-industrial complex. The blockchain's public nature allows participants to verify the history of transactions and records, which helps to prevent potential disputes and conflicts, and hence recourse to legal remedies [9]. Three key advantages of blockchain in the context of increasing the efficiency of blockchain in agro-industrial complex can be recognized as exhaustive [22].

The past five-year period of industry and information technology development continues to influence the conjuncture of the agricultural industry and its individual segments. It should be recognized that the increasing pace of digitalization will steadily stimulate both the market and agro-industrial complexes to further progressive development.

Notably, several major trends have emerged in the agricultural industry, which require the use of modern digital technologies with the widest possible reach. The obligation of market participants to work in Federal State Information System (FSIS) "Zerno", FSIS "Saturn" and the Unified Federal Information System on agricultural lands (EFIS ZSN) (from 2024) has not only contributed to digital transformation, but also to the emergence of a new position in the agro-industrial complex – specialists for reporting in the digital services of the Ministry of Agriculture. The demand for these employees is especially high from enterprises that do not use integrated digital products, which allow uploading some data into FSIS. It is expected that with the growing integration of FMS (agricultural production management systems) with the information systems of the Ministry of Agriculture, the demand for these personnel will decrease, but in 2024 they will still be in demand.

State subsidies to enterprises that use Russian-made machinery, software, seeds and other means are crucial to support the industry. More dynamic development of digital technologies in the agro-industrial complex in 2024 will be stimulated by a draft governmental decree on subsidies for companies that undertake to annually implement solutions included in the unified register of Russian software and databases and using artificial intelligence.

The Russian President has highly assessed the great potential of using drones in the agro-industrial complex [23]. In the future, a special legal regime will be needed for the mass use of digital technologies in agriculture and its sub-sectors. The government is willing to allocate additional state subsidies for producers and industrialists in order to develop the market and generate new profits. Although not all digital technologies in Russia are used as widely as possible, and some are restricted (in particular, the use of drones), the digital technology market is not stagnating; on the contrary, it is developing, and revenues from the use of such technologies in any sphere of human activity are increasing.

In general, the company sees prospects for growth in the rate of digitalization of the agro-industrial complex using digital technologies, including blockchain technology, estimating it in 2024 at the level of 40-50% to the previous year.

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